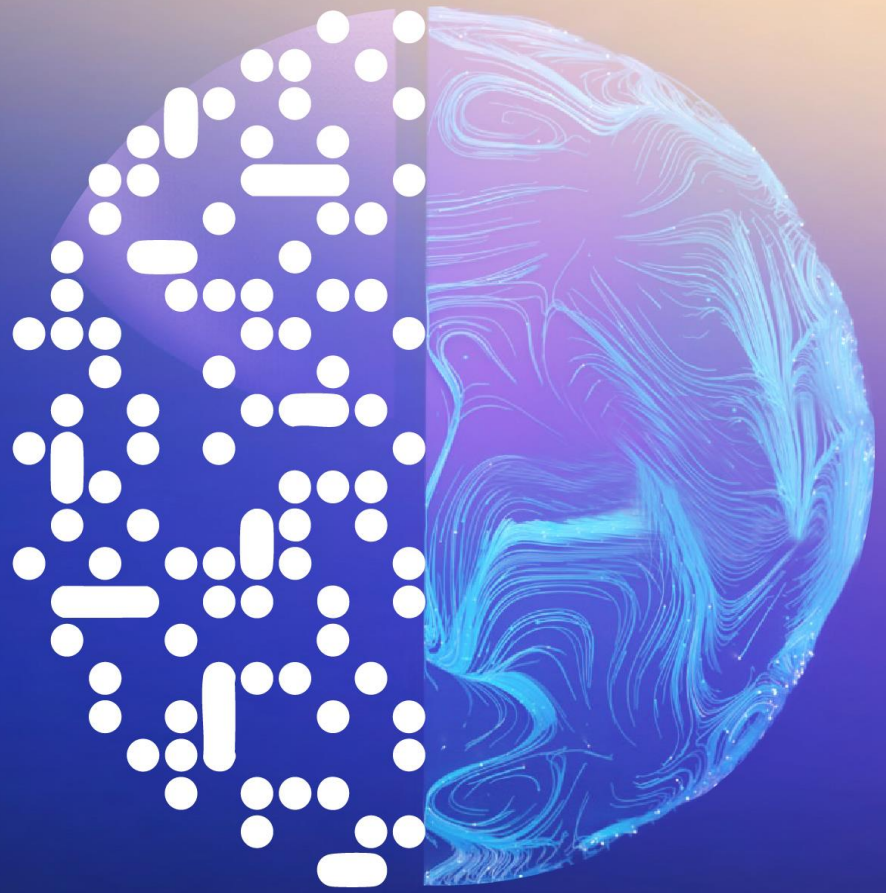
بتنظيم من
Organized by

TDRA | هيئة تنظيم الاتصالات والحكومة الرقمية
TELECOMMUNICATIONS AND DIGITAL GOVERNMENT REGULATORY AUTHORITY

ICTFUND | صندوق الاتصالات وتقنية المعلومات
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Challenges Profiles Overview

Developed and Facilitated by:

Global Innovation & Entrepreneurship

The Failure Point

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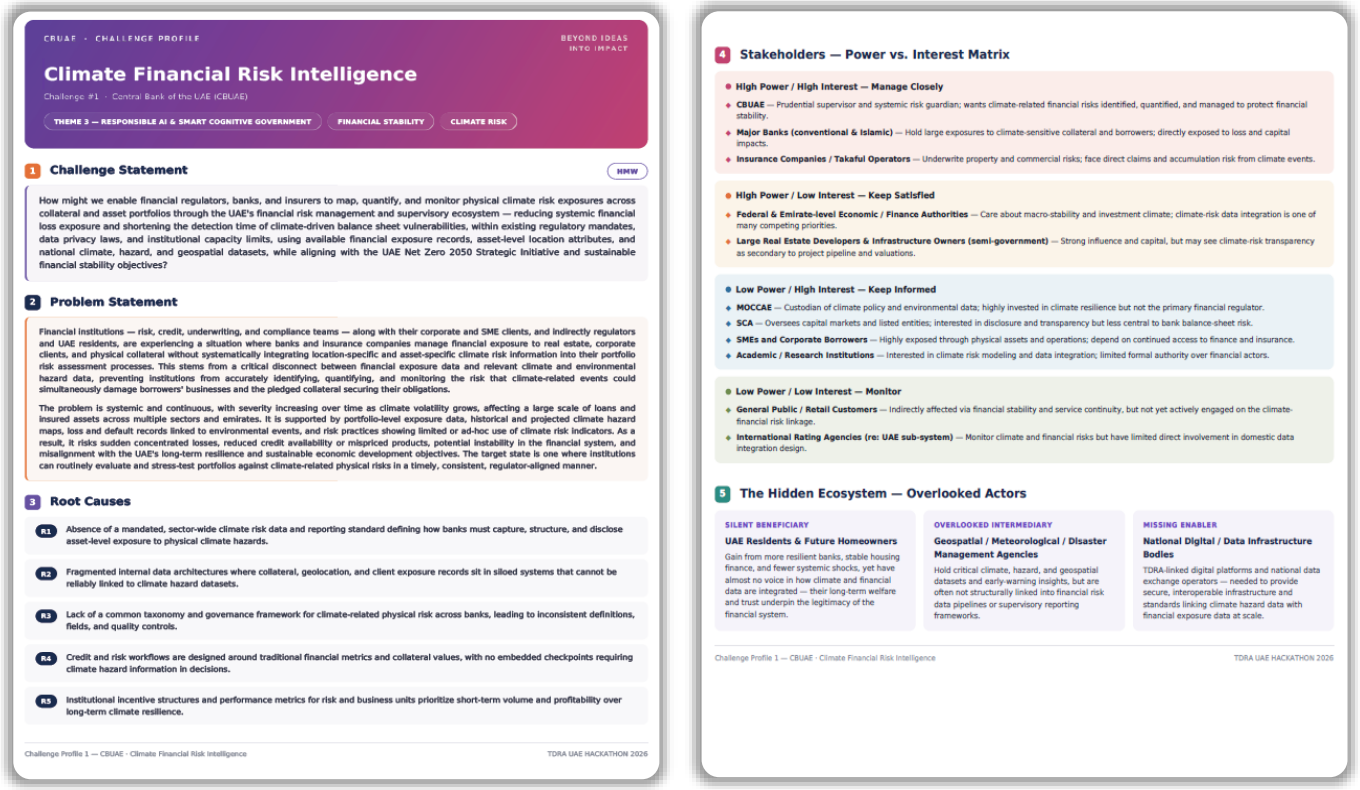


Most government innovation challenges fail not because the problem isn't real, but because the framing prevents innovators from effectively solving it.

What is the Challenge Profile Document?

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- A **structured profile** that defines and analyzes a challenge beyond its mere statement.
- Explores the problem through its **context**, root causes, stakeholders, and ecosystem dynamics.
- Creates a **shared understanding** to guide teams toward solving root causes, not just symptoms.
- Serves as the **primary reference** for participants, mentors, judges, and challenge owners throughout the UAE Hackathon.



CBUAE · CHALLENGE PROFILE
BEYOND IDEAS INTO IMPACT

Climate Financial Risk Intelligence

Challenge #1 · Central Bank of the UAE (CBUAE)

THEME 3 — RESPONSIBLE AI & SMART COGNITIVE GOVERNMENT | FINANCIAL STABILITY | CLIMATE RISK

1 Challenge Statement HMW

How might we enable financial regulators, banks, and insurers to map, quantify, and monitor physical climate risk exposures across collateral and asset portfolios through the UAE's financial risk management and supervisory ecosystem — reducing systemic financial loss exposure and shortening the detection time of climate-driven balance sheet vulnerabilities, within existing regulatory mandates, data privacy laws, and institutional capacity limits, using available financial exposure records, asset-level location attributes, and national climate, hazard, and geospatial datasets, while aligning with the UAE Net Zero 2050 Strategic Initiative and sustainable financial stability objectives?

2 Problem Statement

Financial institutions — risk, credit, underwriting, and compliance teams — along with their corporate and SME clients, and indirectly regulators and UAE residents, are experiencing a situation where banks and insurance companies manage financial exposure to real estate, corporate clients, and physical collateral without systematically integrating location-specific and asset-specific climate risk information into their portfolio risk assessment processes. This stems from a critical disconnect between financial exposure data and relevant climate and environmental hazard data, preventing institutions from accurately identifying, quantifying, and monitoring the risk that climate-related events could simultaneously damage borrowers' businesses and the pledged collateral securing their obligations.

The problem is systemic and continuous, with severity increasing over time as climate volatility grows, affecting a large scale of loans and insured assets across multiple sectors and emirates. It is supported by portfolio-level exposure data, historical and projected climate hazard maps, loss and default records linked to environmental events, and risk practices showing limited or ad-hoc use of climate risk indicators. As a result, it risks sudden concentrated losses, reduced credit availability or mispriced products, potential instability in the financial system, and misalignment with the UAE's long-term resilience and sustainable economic development objectives. The target state is one where institutions can routinely evaluate and stress-test portfolios against climate-related physical risks in a timely, consistent, regulator-aligned manner.

3 Root Causes

R1 Absence of a mandated, sector-wide climate risk data and reporting standard defining how banks must capture, structure, and disclose asset-level exposure to physical climate hazards.

R2 Fragmented internal data architectures where collateral, geolocation, and client exposure records sit in siloed systems that cannot be reliably linked to climate hazard datasets.

R3 Lack of a common taxonomy and governance framework for climate-related physical risk across banks, leading to inconsistent definitions, fields, and quality controls.

R4 Credit and risk workflows are designed around traditional financial metrics and collateral values, with no embedded checkpoints requiring climate hazard information in decisions.

R5 Institutional incentive structures and performance metrics for risk and business units prioritize short-term volume and profitability over long-term climate resilience.

Challenge Profile 1 — CBUAE · Climate Financial Risk Intelligence TDRA UAE HACKATHON 2026

4 Stakeholders — Power vs. Interest Matrix

- High Power / High Interest — Manage Closely
  - **CBUAE** — Prudential supervisor and systemic risk guardian; wants climate-related financial risks identified, quantified, and managed to protect financial stability.
  - **Major Banks (conventional & Islamic)** — Hold large exposures to climate-sensitive collateral and borrowers; directly exposed to loss and capital impacts.
  - **Insurance Companies / Takaful Operators** — Underwrite property and commercial risks; face direct claims and accumulation risk from climate events.
- High Power / Low Interest — Keep Satisfied
  - **Federal & Emirate-level Economic / Finance Authorities** — Care about macro-stability and investment climate; climate-risk data integration is one of many competing priorities.
  - **Large Real Estate Developers & Infrastructure Owners (semi-government)** — Strong influence and capital, but may see climate-risk transparency as secondary to project pipeline and valuations.
- Low Power / High Interest — Keep Informed
  - **MOCCAE** — Custodian of climate policy and environmental data; highly invested in climate resilience but not the primary financial regulator.
  - **SCA** — Oversees capital markets and listed entities; interested in disclosure and transparency but less central to bank balance-sheet risk.
  - **SMEs and Corporate Borrowers** — Highly exposed through physical assets and operations; depend on continued access to finance and insurance.
  - **Academic / Research Institutions** — Interested in climate risk modeling and data integration; limited formal authority over financial actors.
- Low Power / Low Interest — Monitor
  - **General Public / Retail Customers** — Indirectly affected via financial stability and service continuity, but not yet actively engaged on the climate-financial risk linkage.
  - **International Rating Agencies (re: UAE sub-system)** — Monitor climate and financial risks but have limited direct involvement in domestic data integration design.

5 The Hidden Ecosystem — Overlooked Actors

SILENT BENEFICIARY
UAE Residents & Future Homeowners
Gain from more resilient banks, stable housing finance, and fewer systemic shocks, yet have almost no voice in how climate and financial data are integrated — their long-term welfare and trust underpin the legitimacy of the financial system.

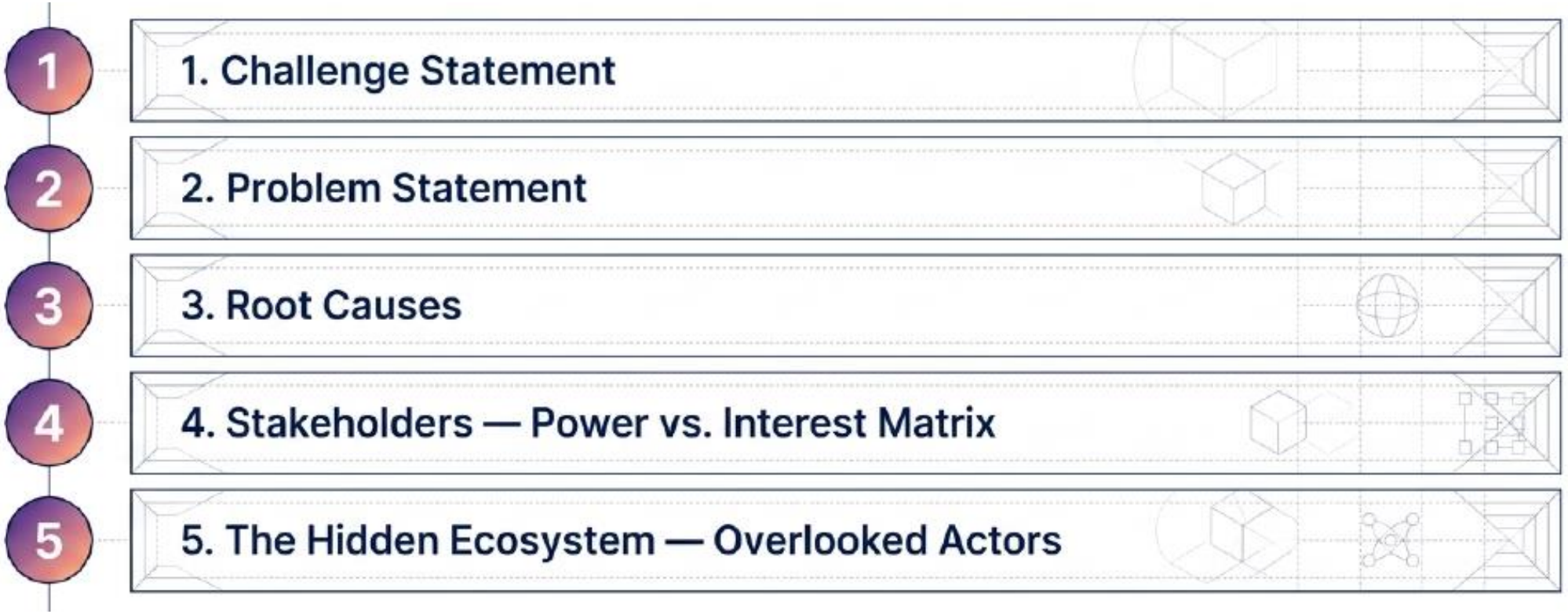
OVERLOOKED INTERMEDIARY
Geospatial / Meteorological / Disaster Management Agencies
Hold critical climate, hazard, and geospatial datasets and early-warning insights, but are often not structurally linked into financial risk data pipelines or supervisory reporting frameworks.

MISSING ENABLER
National Digital / Data Infrastructure Bodies
TDRA-linked digital platforms and national data exchange operators — needed to provide secure, interoperable infrastructure and standards linking climate hazard data with financial exposure data at scale.

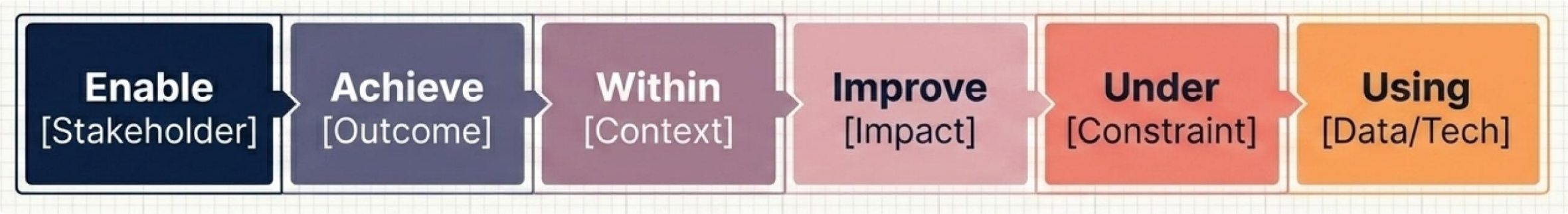
Challenge Profile 1 — CBUAE · Climate Financial Risk Intelligence TDRA UAE HACKATHON 2026

Sample Challenge Profile

Challenge Profile Anatomy



The Challenge Statement Formula



HMW

How might we ..

Enable **[Stakeholders]** to achieve **[Outcome]** within **[Context]** to improve **[Impact]** under **[Constraint]** using **[Data/Tech],** while **[aligning with [National Priority/Strategy/Policy Objective]**

A Challenge Statement Sample

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CBUAE · CHALLENGE PROFILE

BEYOND IDEAS
INTO IMPACT

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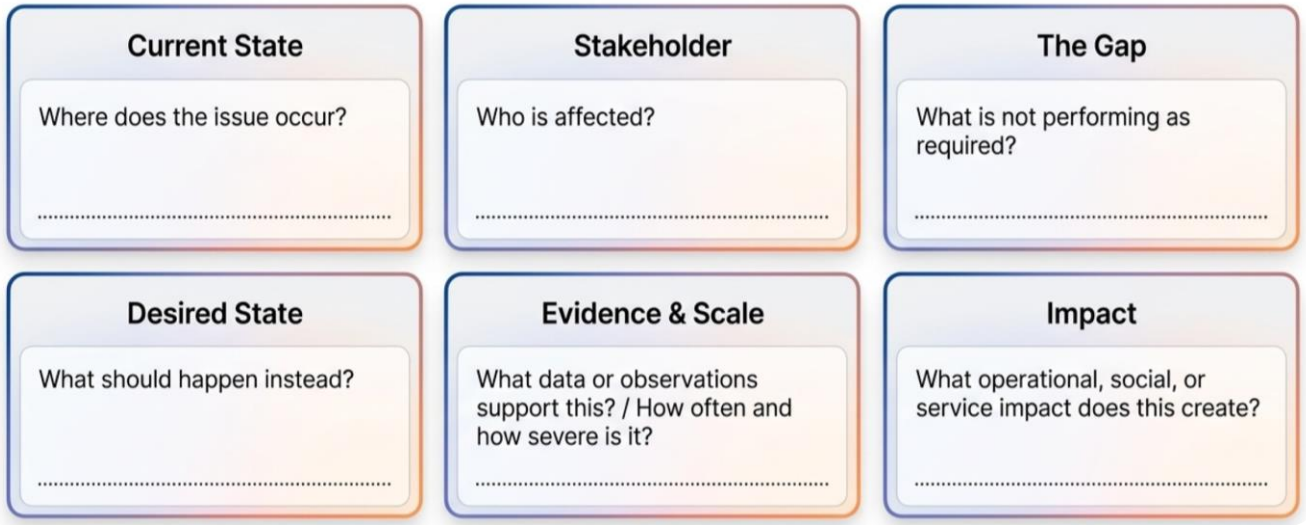
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The Problem Statement Formula



The [**Stakeholders**] are experiencing [**Current State**] due to [**The Gap**]. This problem occurs according to a [**Frequency**] and affects this [**Scale**] and it is supported by [**Evidence**]. As a result, it has this [**Impact**]. It's required to achieve a [**Desired State**].

A Problem Statement Sample

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2 Problem Statement

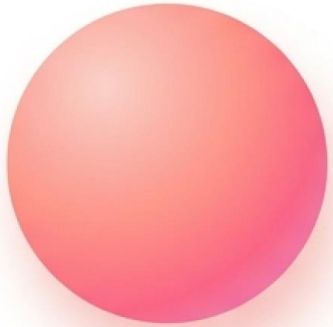
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The Root Causes

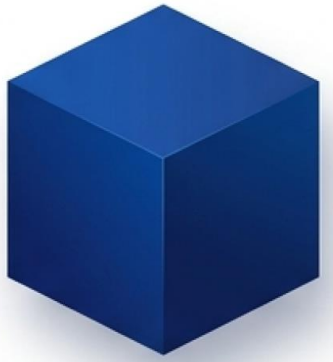
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Most challenges fail for one predictable reason: they address the symptoms, not the root causes



Symptoms.

- Surface-level
- Temporary fixes
- Highly visible
- Reactive



Root Causes.

- Systemic constraints
- Permanent solutions
- Hidden
- Proactive

Root Causes Sample

3 Root Causes

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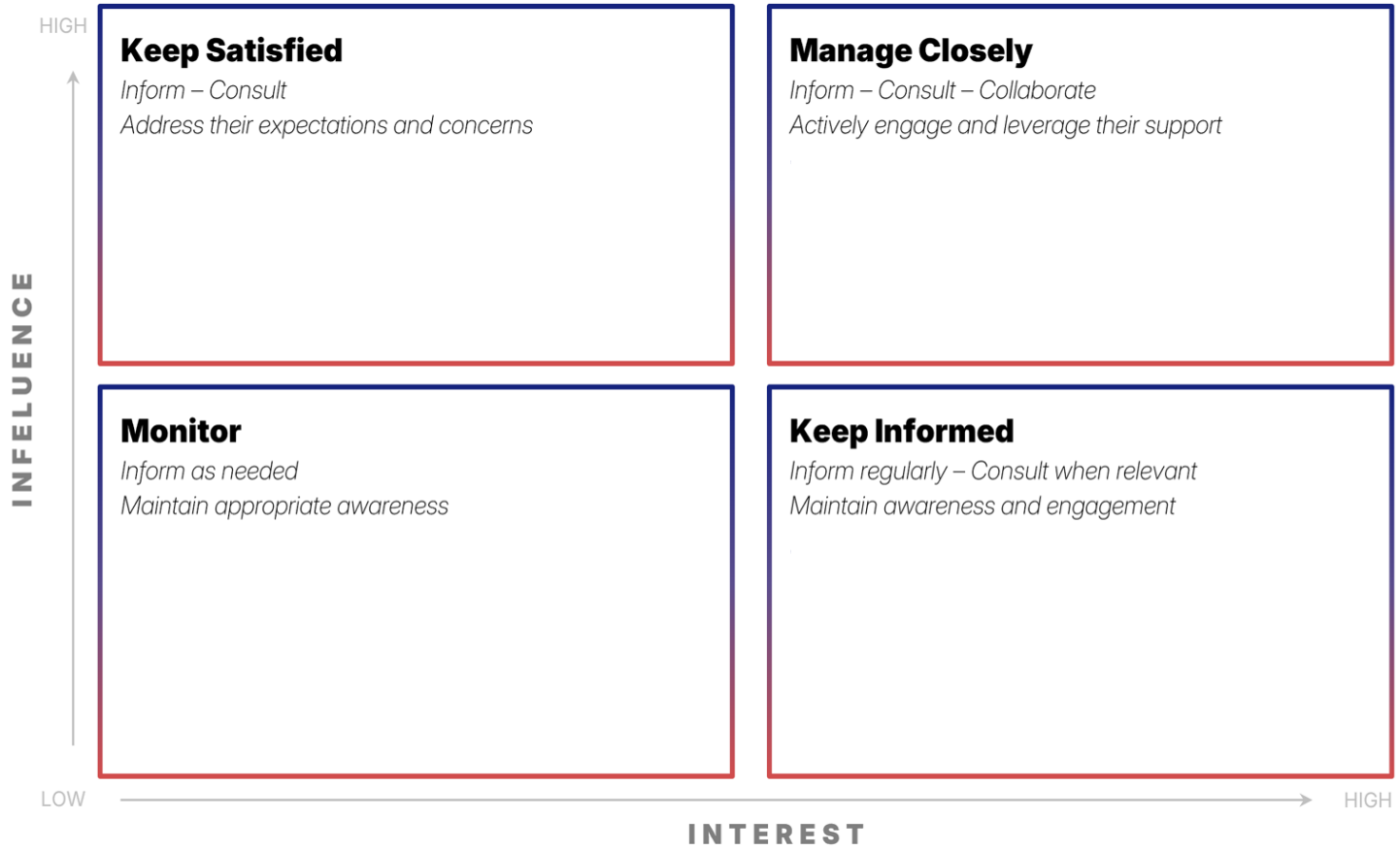
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Stakeholder Mapping

*Mapped as per the **Influence vs Interest** Grid*



Stakeholder Mapping Sample

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The Hidden Ecosystem

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An AI-augmented insight into the following three categories in the systems perspective view of the challenge statement

1

SILENT BENEFICIARY

Who benefits but is missing

2

OVERLOOKED INTERMEDIARY

Key execution actor missing

3

MISSING ENABLER

External capability/actor required for success

The Hidden Ecosystem Sample

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Thank You ..

